2ND SEMESTER LAB REPORT

TITLE : Electrical Circuit Analysis Lab



Faiq Ali (249520)

Muhammad Zeeshan (243346)

Muhammad Waqas (240756)

BACHELOR OF COMPUTER ENGINEERING (Fall-2025)

Submitted To:

Engr. Shanzay Jamil

Lab Engineer

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING FACULTY OF ENGINEERING AIR UNIVERSITY, ISLAMABAD

AIR UNIVERSITY

DEPARTMENT OF ELECTRICAL AND COMPUTER ENGINEERING

EXPERIMENT NO 10

Lab Title: IMPLEMENTATION AND VERIFICATION OF THEVENIN THEOREM BOTH ON HARDWARE AND SOFTWARE

Objectiv: THEVENIN THEOREM

Student Name: Muhammad Waqas -Muhammad Zeesha -Faiq Ali

LAB ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Ability to Conduct Experiment					
Ability to assimilate the results					
Effective use of lab equipment and follows the lab safety rules					

Total Marks: _____ Obtained Marks: _____

LAB REPORT ASSESSMENT:

Attributes	Excellent (5)	Good (4)	Average (3)	Satisfactory (2)	Unsatisfactory (1)
Data presentation					
Experimental results					
Conclusion					

Total Marks: _____ Obtained Marks:

Date: 8/May/2025 Signature:

LABORATORY EXPERIMENT NO. 10

IMPLEMENTATION AND VERIFICATION OF THEVENIN THEOREM BOTH ON HARDWARE AND SOFTWARE

OBJECTIVE:

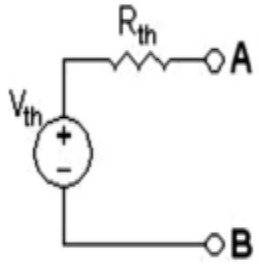
1. To find experimentally the values of components required to construct a Thévenin's equivalent circuit.
2. To check the experimental values versus calculated values.
3. To check the validity of Thévenin's theorem.
4. To verify the results through software using PSpice / Proteus..

Thévenin Theorem

As mentioned in the previous lab that the last three labs will be related to network theorems. So, in this lab we will be practically implementing another network theorem known as the "Thévenin Theorem". Thévenin's Theorem allows one to replace a complicated circuit with a simple equivalent circuit containing only a voltage source and a series connected resistor. The theorem is very important
from both theoretical and practical point of views. Concisely stated, Thévenin's

Theorem says:

"Any two-terminal linear circuit can be replaced by an equivalent circuit consisting of a voltage source (V th) and a series resistor (R th)"



Lab Tasks:

Lab task 1:

d) Measure the value of V L and I L using a DMM. Record these values and compare them with the theoretical values of V L and I L.

VL = 6.11V **IL = 0.90mA**

f) Turn on the trainer and using a DMM, measure the voltage across the terminals A and B i.e. the open-circuit voltage Voc or Vth. Record this value and compare it with the theoretical value of Vth.

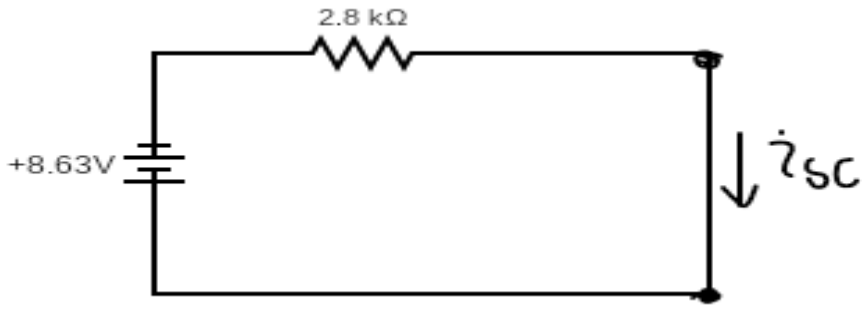
Vth = 8.63

Also measure the short-circuit current, I sc by placing the DMM directly between terminals A and B, as shown. Record this value and compare it with the theoretical value of I sc.

Isc = 3.09mA

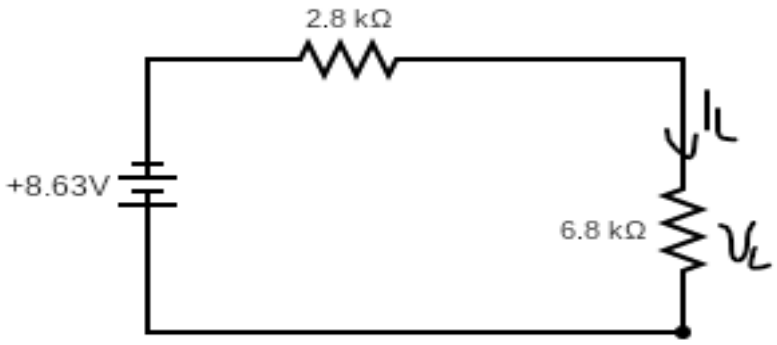
I) Turn on the trainer and measure the short circuit current I sc between terminals A and B. this value should be the same as the one calculated in step (g) and also the same as the theoretical value of Isc.

Isc = 3.08



n) Turn on the trainer and measure V L , the voltage across load, as Valus are totally same to the original circuit

VL =6.11 V **IL = 0.90mA**



Theoretical Calculation

Thévenin Analysis

Step 1: Calculate VTH

VTH = Va – Vb

Va = 12V × 10kΩ/(3.9kΩ + 10kΩ) = 12V × 10/13.9 = 8.63V

Vb = 0V VTH = 8.63V – 0

Thévenin VTH = 8.63V

Step 2: Calculate RTH

R3 || R4 = (3.9kΩ × 10kΩ)/(3.9kΩ + 10kΩ)

= 39000/13.9 = 2.81kΩ RTH = 10kΩ + 2.81kΩ = 12.81kΩ

Step 3: Calculate Voltage Across R2

I = VTH/(RTH + R2) = 8.63V/(12.81kΩ + 6.8kΩ)

= 8.63V/19.61kΩ = 0.44mA VR2 = I × R2 = 0.44mA × 6.8kΩ = 2.99V

Corrected Analysis

VTH = 8.63V I = 8.63V/RTH = 8.63V/12.81kΩ

= 0.674mA VR2 = 8.63V × 10kΩ/(10kΩ + 6.8kΩ)

= 8.63V × 10/16.8 = 5.14V

Circuit Configuration Consideration

R1 || R2 = (10kΩ × 6.8kΩ)/(10kΩ + 6.8kΩ) = 68000/16.8 = 4.05kΩ

RT = 4.05kΩ + 3.9kΩ + 10kΩ = 17.95kΩ

IT = 12V/17.95kΩ = 0.669mA

IR2 = IT × R1/(R1 + R2) = 0.669mA × 10kΩ/16.8kΩ = 0.398mA

VR2 = IR2 × R2 = 0.398mA × 6.8kΩ = 2.71V

Nodal Analysis

(Va-12V)/3.9kΩ + (Va-Vb)/6.8kΩ = 0

(Vb-Va)/6.8kΩ + Vb/10kΩ = 0

Vb = Va × 10kΩ/(10kΩ+6.8kΩ) = 0.595

Va (Va-12V)/3.9kΩ + (Va-0.595Va)/6.8kΩ = 0

(Va-12V)/3.9kΩ + 0.405Va/6.8kΩ = 0

Va = 10.74V Vb = 0.595 × 10.74V = 6.39V

VR2 = Va - Vb = 10.74V - 6.39V = 4.35V

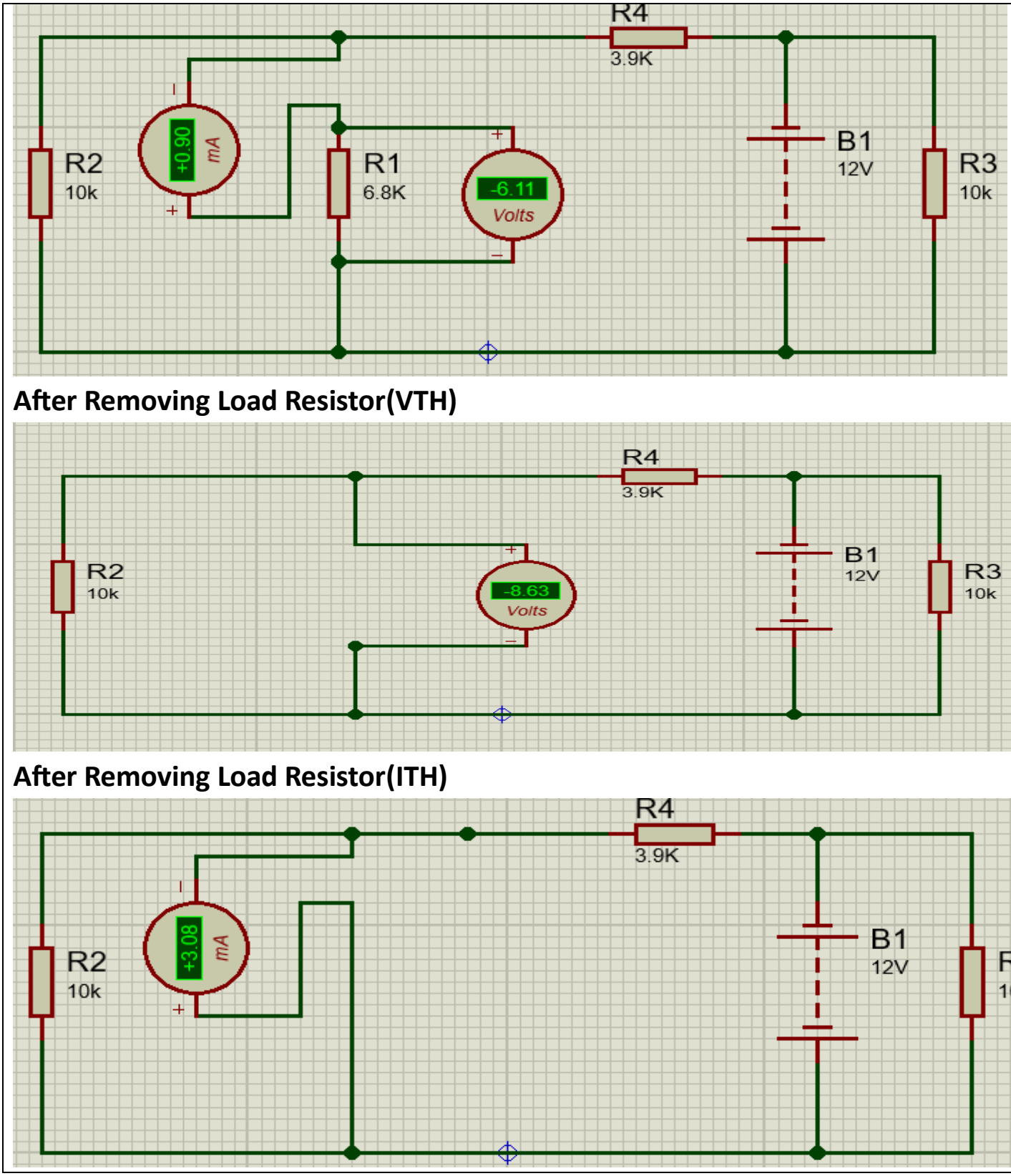
Mesh Analysis

$10k\Omega \times I_1 + 6.8k\Omega \times (I_1-I_2) = 0$ $6.8k\Omega \times (I_2-I_1) + 3.9k\Omega \times I_2 + 12V = 0$ $16.8k\Omega \times I_1 - 6.8k\Omega \times I_2 = 0$ $-6.8k\Omega \times I_1 + 10.7k\Omega \times I_2 = -12V$ $I_1 = 0.4mA$, $I_2 = 0.9mA$ $IR2 = I_2 - I_1 = 0.9mA - 0.4mA = 0.5mA$

VR2 = 0.5mA × 6.8kΩ = 3.4V

Simulation result: VR2 = 6.11V

Protius Simulation



After Removing Load Resistor(VTH)

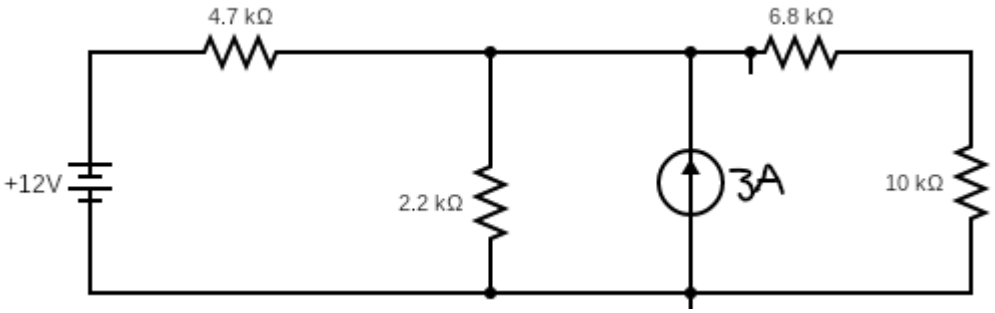
After Removing Load Resistor(ITH)

Lab Task 2:

2. Solve the following circuit by Thevenin Theorem:

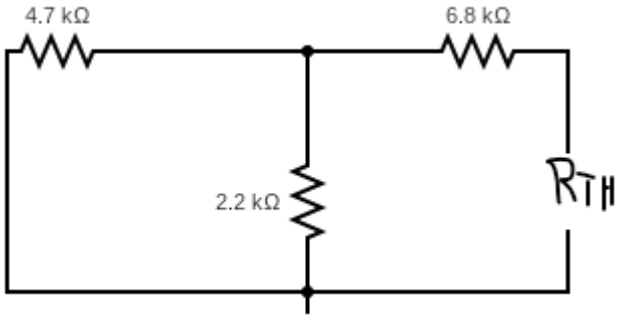
For the circuit explained and solved in the manual earlier, do the calculation yourselves and simulate it on Proteus. Also, perform the task on hardware and verify your results. Before performing any of the tasks prior mentioned, modify the circuit such that remove the current source, the 25V source is equal to 12V, the resistors of 5Ω = 4.7kΩ, 20Ω = 2.2kΩ, 4Ω = 6.8kΩ and 24Ω = 10kΩ. Redraw the modified circuit and the write down step-by-step procedure with diagrams (as given in the previous task).

CIRCUIT DIAGRAM



The portion of the circuit outlined is to be replaced by it Thévenin equivalent circuit.

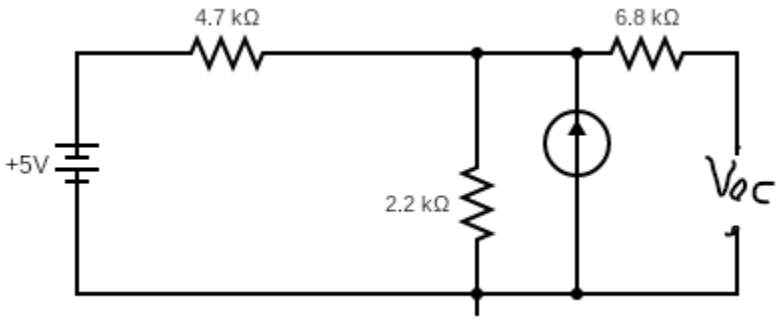
To determine R th, set all sources to zero by replacing voltage source by short-circuit and current source by an open circuit, as shown:



Rth=(4700‖2200)+6800

Rth ≈ 8298.55Ω

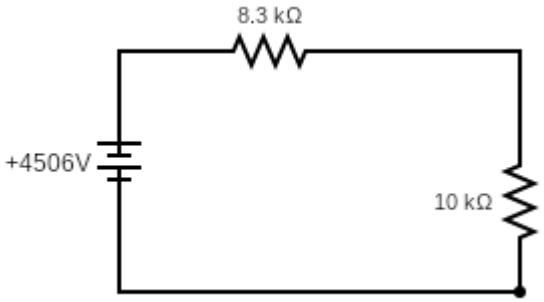
Thévenin Equivalent VOLTAGE:



(v - 12) / 4700 + v / 2200 = 3

VTH ≈ 4499.48

Thévenin Equivalent Circuit:



LOAD VOLTAGE

V_L ≈ 2513.63 V

LOAD CURRENT

I_L ≈ 0.251365 A

Theoretical Calculation

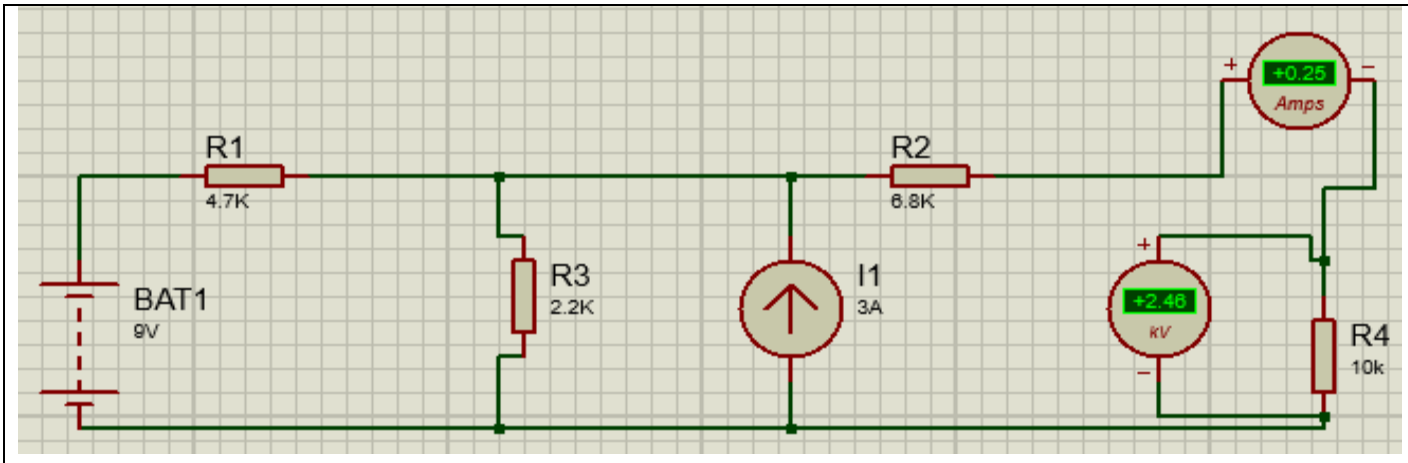
Rth = (4700‖2200) + 6800
First, I need to calculate the parallel combination of 4700Ω and 2200Ω:
4700‖2200 = (4700 × 2200)/(4700 + 2200) = 10340000/6900 = 1498.55Ω
Now, I add the series resistor:
Rth = 1498.55 + 6800 = 8298.55Ω
Therefore,
Rth ≈ 8298.55Ω

(v - 12) / 4700 + v / 2200 = 3
103400 * (VTH - 12) / 4700 + 103400 * VTH / 2200 = 3 * 103400
22 * (VTH - 12) + 47 * VTH = 310200
22VTH - 264 + 47VTH = 310200
69VTH - 264 = 310200
69VTH = 310200 + 264
69VTH = 310464
VTH = 310464 / 69
VTH ≈ 4499.48

V_L = (R_L / (R_th + R_L)) * V_th V_L = (10000 / (8298 + 10000)) * 4599.48
V_L = (10000 / 18298) * 4599.48
V_L ≈ 0.546507815 * 4599.48
LOAD VOLTAGE
V_L ≈ 2513.63 V

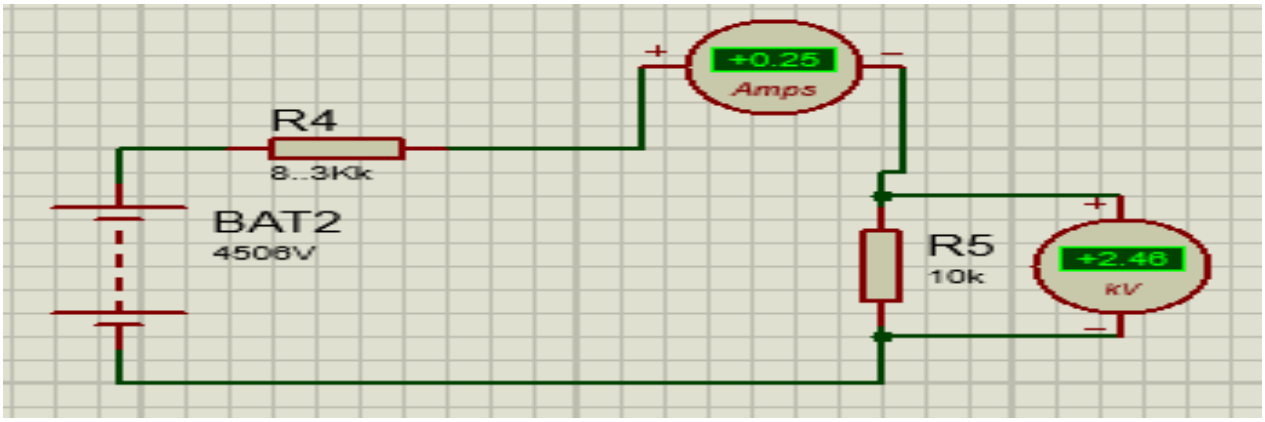
I_L = V_th / (R_th + R_L) I_L = 4599.48 / (8298 + 10000)
I_L = 4599.48 / 18298
LOAD CURRENT
I_L ≈ 0.251365 A

Protius Simulation



Thévenin Equivalent Circuit:

As values are same in original circuit and in final Thévenin Equivalent Circuit:



Conclusions:

After performing the Thévenin circuit problems, we conclude that Thévenin's Theorem is a powerful tool that helps us simplify complex electrical circuits into a much simpler form. By converting a network into a single voltage source (V□□) in series with a resistor (R□□), it becomes much easier to analyze how a circuit behaves, especially at two specific terminals. This simplification helps us focus only on the part of the circuit we're interested in, without dealing with the entire original network. It is especially useful for analyzing circuits with varying loads and for reducing calculation time. Overall, Thévenin's Theorem makes solving and understanding electrical circuits much more manageable.